Lab/Project Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	7					
Implementation	8					
Report writing	10					
					Total obtained mark	
Comments						

Semester: Fall 2026

Student Name: Ashikur Rahman Sazim, Prarthona Biswas puja, Sabiha nazir.

Student ID:241-35-034, 241-35-187, 241-35-279

Batch: 42 **Section:B1**

Course Code: SE331 **Course Name:** Software Engineering Design CapstoneProject

Course Teacher Name: Rahat Uddin Azad

Designation: Lecturer

Submission Date: 3/10/26

PROJECT PROPOSAL REPORT

Statute-Grounded Bangla Legal Assistant

A Retrieval-Augmented Generation (RAG) system that answers legal questions in Bangla, cites the exact statute and section, analyses user-provided documents, and helps with drafting and next steps

Course	SE331 - Software Engineering Design Capstone Project
Instructor	Rahat Uddin Azad, Lecturer
Group No. / Name	Group No. 07
Team Members	Ashikur Rahman Sazim (241-35-034); Prarthona Biswas Puja (241-35-187); Sabiha Nazir (241-35-279)
Semester / Batch / Section	Fall 2026 / Batch 42 / Section B
Submission Date	3/10/26

Table of Contents

1. Introduction

- 1.1 Background Overview
- 1.2 Problem Statement
- 1.3 Objectives
- 1.4 Scope
- 1.5 Stakeholders
- 1.6 Proposed Solution
- 1.7 AI Features

2. System Requirements

- 2.1 Functional Requirements
- 2.2 AI Requirements
- 2.3 Non-Functional Requirements

3. Tools and Technologies

4. Project Timeline and Work Plan

- 4.1 Milestone Table
- 4.2 Task Distribution Among Team Members

5. Additional Sections

- 5.1 Existing System
- 5.2 System Architecture
- 5.3 Ethical, Legal, and Social Considerations
- 5.4 Future Work

1. Introduction

1.1 Background Overview

Application domain. The application domain of this project is Legal Technology (LegalTech), specifically legal information access and public legal awareness in Bangladesh. In Bangladesh, laws are written as Acts, and each Act is divided into numbered sections. Official consolidated texts of the Acts are published online by the Legislative and Parliamentary Affairs Division through the Bangladesh Laws portal (bdlaws.minlaw.gov.bd). Many Acts are available in Bangla, while older laws such as the Penal Code, 1860 are mainly available in English.

Current practice. When an ordinary person wants to know what the law says about a situation (for example, a worker whose employer has not paid wages, or a consumer who was sold an expired product), the usual options are: asking a lawyer, asking friends or relatives, searching Facebook groups or general web pages, or reading the Act directly. A lawyer's time costs money and not everybody can afford it. Reading an Act directly is difficult because legal Bangla is formal, one situation may involve several sections spread across different Acts, and the portal supports only plain keyword search, which fails when the user describes the problem in everyday words. People also hold documents (appointment letters, notices, termination letters, receipts, complaint drafts) that they cannot easily check against the law, and they often do not know what to write in a complaint or where to go next.

Use of AI in this domain. General-purpose chatbots (such as ChatGPT or Gemini) can answer legal questions in Bangla, but they are not tied to the actual text of Bangladeshi statutes. They may quote a section number that does not exist, mix Bangladeshi law with Indian or Pakistani law, or use an outdated or repealed version, and they usually cannot show the exact legal text behind an answer. A few local legal-information websites exist, but they are mostly static article collections. Therefore, a Bangla-first assistant that is grounded in the official statute text, always shows its source, and can also work on the user's own documents is still largely missing.

1.2 Problem Statement

Problem. Ordinary citizens cannot easily find which law and which section applies to their situation, cannot easily check their own documents against the law, and do not know how to prepare a complaint or what step to take next. The reasons are: (a) statutes are long and written in formal legal language; (b) keyword search on the official portal does not understand questions written in everyday Bangla; (c) general AI chatbots give fluent answers that are not verifiably based on Bangladeshi statutes and may be wrong; and (d) no easy tool exists to check whether a document or a described case is supported by, or inconsistent with, the statute text.

Gaps in existing systems.

- The official portal supports lookup by Act name and keyword but cannot answer a natural-language question or read a user's document.

- General chatbots can hallucinate section numbers or content and do not reveal their source or how well-supported an answer is.
- Most existing tools are English-first; Bangla questions and Bangla documents are handled less reliably.
- Existing sources rarely distinguish between 'this is what the statute says' and 'this is advice', and rarely tell the user which information or evidence is missing from their case.
- No freely available tool combines question answering, document checking, drafting help, and next-step guidance for Bangladeshi statutes in Bangla.

Who is affected. Members of the general public (workers, consumers, women facing domestic problems, internet users facing online harassment), law students who need to locate provisions quickly, and community organisations or legal-aid volunteers who answer basic legal queries repeatedly.

Why software is needed. A software system can be available at any time, can search thousands of sections in seconds, and can present the original legal text next to a simple explanation. It can also read a document and compare it with many sections at once, which is not practical to do manually for every person.

Why AI is needed. Today, a person must manually read through Acts, decide which section may be relevant, interpret the formal language, compare their own papers with the law, and work out what is missing. AI can automate exactly these steps: (1) semantic search matches a question written in everyday Bangla to the relevant sections even when no words are shared; (2) a language model rewrites the retrieved legal text into a short plain-Bangla explanation; (3) language-model-based extraction identifies the legal issue, the key facts, claims, dates and amounts in a question or document; and (4) these extracted items are compared with retrieved sections to flag consistency, contradictions and missing information. Rule-based or keyword approaches cannot do this reliably because the same situation can be described in many different ways.

Scope realism. The project is intentionally limited to a fixed set of selected Acts, to text-based input, and to legal information. It does not claim to give legal advice, to decide whether a document is genuine, to establish whether the facts told by a user are true, to predict court outcomes, or to cover case law. All analysis features compare the user's text only against the stored statute snapshot.

1.3 Objectives

General Objective. To develop a Bangla-language legal information assistant that helps citizens understand which provisions of selected Bangladeshi statutes relate to their situation, by answering questions in simple Bangla with exact Act and section references, analysing the documents and case descriptions they provide, helping them prepare a basic Bangla draft, and guiding them on the information and steps that the statutes support.

AI Objective. To build a Retrieval-Augmented Generation (RAG) pipeline and a set of analysis modules that (i) retrieve the most relevant statute sections for a Bangla question or document using multilingual semantic search; (ii) generate answers that are based only on the retrieved sections and cite them; (iii) verify every citation and show an evidence-status indicator for every output; (iv) extract legal issues, facts and

claims from questions and uploaded documents and check them for consistency with the statutes, internal contradictions and missing information; (v) produce template-based Bangla drafts and statute-supported next-step guidance; (vi) map the facts of a case to statute sections and to the evidence provided, build an evidence matrix, and generate a structured case summary; and (vii) decline to answer when the knowledge base does not contain sufficient support. The system will be evaluated on a self-prepared Bangla test set for retrieval accuracy, answer faithfulness, and the quality of each analysis feature.

1.4 Scope

In scope

- A knowledge base built from the official text of a limited set of Acts (initially four to five; proposed: Bangladesh Labour Act 2006, Consumer Rights Protection Act 2009, Cyber Security Act 2023, Domestic Violence (Prevention and Protection) Act 2010, and selected chapters of the Penal Code 1860). The final list will be fixed in the first two weeks according to text availability and workload.
- Question input in Bangla (typed text; Banglish/English questions handled on a best-effort basis).
- Core AI (F1-F4): Bangla legal Q&A, statute-grounded RAG, exact section citation with source verification, and legal issue extraction.
- Document and case intelligence (F5-F8): upload or paste of text-based documents (typed PDF, DOCX, TXT, or pasted text; limited length) and analysis, a verification engine that compares claims and clauses with retrieved statute sections, a contradiction detector, and a missing-information / evidence detector.
- Legal assistance (F9-F10): template-based Bangla drafting for a small set of common documents (for example a complaint or application letter) and a next-step guide based on procedures found in the selected Acts.
- Trust layer (F11-F12): an evidence-status indicator (supported / partly supported / not supported) shown with every output, and a finder for related provisions and similar example scenarios from a curated scenario library prepared by the team.
- Case analysis and reporting (F13-F15): a mapping from the facts of a case to the applicable statute sections and the evidence given by the user, an evidence matrix that shows which evidence supports or is missing for each point, and a structured Bangla case summary that can be shown or downloaded.
- A 'not enough information' response when retrieval confidence is low.
- A simple web interface, an answer feedback button (helpful / not helpful), and a mandatory legal-information disclaimer.
- An evaluation module with a test set of at least 100 questions plus a smaller set of sample documents (about 30) for the document features.

Out of scope

- Legal advice, filing documents with any authority, or predicting the outcome of a case.
- Authenticating documents (detecting forgery or fake signatures) or deciding whether the facts told by a user are true. 'Verification' in this project means comparison with the statute text only.

- Court judgments, case law, rules and regulations, gazette notifications, and all Acts not in the selected list. The 'similar case' feature uses curated example scenarios, not real court cases.
- OCR of scanned or handwritten documents, voice input, and mobile app development.
- Judging the strength, admissibility or genuineness of evidence; the evidence matrix only records what the user has provided and what the statute text or checklist mentions.
- Fully automatic legal documents that are ready to file; all drafts are templates that the user must review.
- Automatically tracking amendments in real time (the knowledge base is a fixed snapshot, with its date shown).
- User accounts with personal case files; uploaded documents are processed in the session and are not stored permanently.

What the AI will do: understand the question or document, extract the legal issue and key facts, find relevant sections, summarise them in simple Bangla, cite them, compare claims with the statute text, flag possible contradictions and missing information, produce a basic draft from a template, list statute-supported next steps, show an evidence status, and refuse when unsupported. **What the AI will not do:** decide who is legally right or wrong, judge the truth of a document or of the user's facts, recommend legal action, or answer from knowledge outside the retrieved statute text.

System boundary. The system takes a text question and, optionally, a text-based document as input and returns outputs based only on the stored statute snapshot and the user's own text. Anything requiring a lawyer's judgement is outside the boundary and the user is directed to a qualified lawyer or legal-aid service. To avoid scope creep, features are built in three priority tiers (see Section 4.1): Tier 1 core features are completed first, and Tier 3 features are reduced in depth if time runs short.

1.5 Stakeholders

Stakeholder	Role	Interest in the system
General citizens (end users)	Ask questions, upload documents, read answers, create drafts	Quick, understandable, trustworthy information in Bangla at no cost
Law students and teachers	Use as a lookup and learning tool	Fast navigation to correct sections with exact text and related provisions
Legal-aid volunteers / NGOs	Use for first-level information and draft preparation	Reduce repetitive basic queries; verify source text easily; get checklists of missing information
Lawyers (advisors / reviewers)	Review sample answers, drafts and checklists (optional)	Ensure information is not misleading
System administrator (project team)	Maintain knowledge base, models, scenario library, templates, and hosting	Reliable operation, updating statute data, monitoring feedback

Stakeholder	Role	Interest in the system
Course instructor / university	Evaluate the project	Feasible scope, sound AI design, proper evaluation
Legislative authority (data source)	Publisher of official statute text	Correct and fair use of the official text with attribution

1.6 Proposed Solution

The proposed solution is a web-based assistant built on a RAG pipeline. Instead of letting a language model answer from its own memory, the system first retrieves the relevant sections from the official statute text and then asks the model to answer using only those sections. The same retrieved evidence is reused by the document-analysis, drafting and guidance features, so all fifteen features share one pipeline rather than being separate large modules.

Key system features

Group	Feature	What it does
Core AI	F1. Bangla Legal Q&A	Accepts a question in Bangla and returns a short plain-Bangla answer
	F2. Statute-Grounded RAG	Retrieves top-k relevant sections and restricts the answer to them
	F3. Exact Section Citation & Source Verification	Shows Act, section and original text; blocks answers whose citations are not in the retrieved set
	F4. Legal Issue Extraction	Identifies the legal issue category, key facts and likely Act from the input
Document & Case Intelligence	F5. Document Upload & Analysis	Reads a text-based document and summarises its key clauses, dates, amounts and claims in Bangla
	F6. Case & Document Verification Engine	Compares each claim or clause with retrieved sections: consistent / conflicts / not covered
	F7. Contradiction Detector	Flags possible contradictions inside a document or case description, and between it and the statute
	F8. Missing Information / Evidence Detector	Lists information or papers that are commonly needed for the identified issue but are not present
Legal Assistance	F9. Bangla Legal Drafting	Fills a Bangla template (complaint / application / notice) from the user's details, with cited sections
	F10. Legal Procedure / Next-Step Guide	Lists next steps that are stated in the selected Acts, otherwise refers the user to legal aid
Trust Layer	F11. Evidence Status / Support Indicator	Shows supported / partly supported / not supported for each output

Group	Feature	What it does
	F12. Similar Case / Provision Finder	Finds related sections across Acts and similar example scenarios from the curated library
Case Analysis & Reporting	F13. Fact → Law → Evidence Mapping	Links each extracted fact to the applicable statute section and to the evidence the user has provided
	F14. Case Evidence Matrix	Shows a table of legal points against evidence items: provided / missing / unclear
	F15. Case Summary Generator	Produces a one-page structured Bangla summary of the case with sections, evidence status and missing items

Workflow

1. Preparation (offline). Collect the official text of the selected Acts, clean it, and split it section by section, keeping metadata (Act, chapter, section number, title). Convert each section into an embedding vector and store it in a vector database. Also prepare the draft templates, issue checklists and the scenario library. [AI: embedding model]
2. Input. The user types a Bangla question, or uploads / pastes a document or case description.
3. Issue extraction. The system extracts the legal issue category, key facts, claims, dates and amounts, and uses the issue to guide retrieval. [AI: LLM structured extraction] (F4)
4. Retrieval. The question or extracted claims are embedded and the most similar sections are retrieved (top-k), then optionally re-ranked. [AI: semantic search] (F2, F12)
5. Generation and analysis. A language model receives the input and only the retrieved sections and writes the answer, or performs the claim comparison, contradiction check and missing-information check. [AI: LLM] (F1, F5-F8)
6. Drafting and guidance. On request, the system fills a template or lists next steps drawn only from the retrieved text. [AI: LLM] (F9, F10)
7. Case mapping and reporting. For a case or document, the extracted facts are linked to the retrieved sections and to the evidence the user provided, an evidence matrix is built, and a structured case summary is generated. [AI: LLM + retrieval] (F13, F14, F15)
8. Verification. The system checks that every cited section exists in the retrieved set and computes the evidence status. If retrieval confidence is low or the check fails, it replies that it cannot find a supported answer. [AI-assisted check] (F3, F11)
9. Display. The output, the cited Acts and sections, the original text, the evidence status, the snapshot date and a disclaimer are shown. The user can give feedback.

System overview diagram

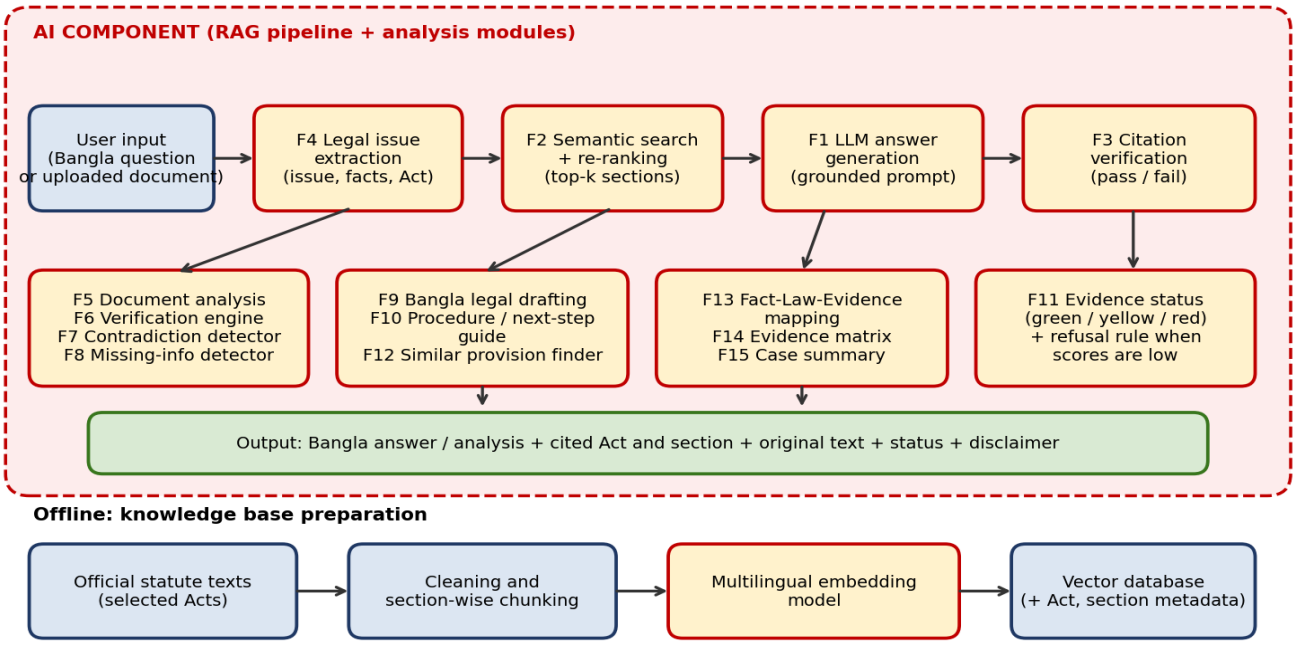


Figure 1: System overview. Components marked in red and yellow are AI components.

1.7 AI Features

Existing AI features and their limitations

Existing feature	Limitation
General LLM chatbots (ChatGPT, Gemini, etc.) answer legal questions in many languages	Not grounded in Bangladeshi statutes; may invent sections or mix jurisdictions; no verifiable source; no indication of how well-supported an answer is; may be outdated
General chatbots with file upload can summarise a document	Do not compare the document with the exact statute text; no section-level evidence; may miss contradictions or invent legal claims
Keyword search on the official laws portal	Needs the user to know the Act and legal terms; no understanding of everyday Bangla; no explanation
Generic English legal-research and drafting tools	Cover other countries' laws; English-first; mostly paid

Proposed AI features

Feature	Input	Output	Manual work replaced
F1. Bangla Legal Q&A	Bangla question (everyday wording)	Plain-Bangla answer (up to about 200 words)	Interpreting formal legal language

Feature	Input	Output	Manual work replaced
F2. Statute-Grounded RAG	Question or extracted claims	Ranked list of relevant sections with scores	Reading Acts and guessing keywords
F3. Exact Section Citation & Source Verification	Answer citations + retrieved sections	Act, section and original text; pass/fail flag; answer blocked if citations are invalid	Manually checking each cited section
F4. Legal Issue Extraction	Question or document text	Issue category, key facts, parties, dates, amounts, likely Act (structured)	Reading a story and deciding which area of law it belongs to
F5. Document Upload & Analysis	Text-based PDF / DOCX / TXT or pasted text	Bangla summary of key clauses, dates, amounts and claims	Reading long documents line by line
F6. Case & Document Verification Engine	Extracted claims / clauses + retrieved sections	Per-claim label: consistent / conflicts with / not covered, with cited section	Comparing a document with the law by hand
F7. Contradiction Detector	Extracted facts, dates, amounts; retrieved sections	List of possible contradictions with the conflicting text shown side by side	Noticing inconsistencies in dates, amounts or statements
F8. Missing Information / Evidence Detector	Identified issue + provided text	Checklist of items present and items not found (e.g. appointment letter, payslip, receipt)	Knowing what papers and details are needed
F9. Bangla Legal Drafting	Issue, user details, retrieved sections, chosen template	Editable Bangla draft with cited sections and placeholders for missing details	Writing a formal letter or application from scratch
F10. Legal Procedure / Next-Step Guide	Identified issue + retrieved sections	Ordered list of steps stated in the Acts, or a referral to legal aid when not stated	Searching for where and how to complain
F11. Evidence Status / Support Indicator	Retrieval scores, verification result, number of supporting sections	Supported / partly supported / not supported label with a short reason	A person judging if the law actually covers the question
F12. Similar Case / Provision Finder	Question, section or scenario	Related sections across Acts and up to 3 similar example scenarios from the curated library	Searching several Acts for related provisions
F13. Fact → Law → Evidence Mapping	Extracted facts (F4/F5) + retrieved sections + user-provided evidence list	Table: fact, applicable section (with citation), evidence provided / needed	Linking each fact to a law and to a proof by hand

Feature	Input	Output	Manual work replaced
F14. Case Evidence Matrix	F13 mapping + documents or evidence items listed by the user	Matrix of legal points vs. evidence items with provided / missing / unclear cells	Building an evidence table or checklist manually
F15. Case Summary Generator	Case description or document + F4, F6, F8, F13 outputs	One-page Bangla summary: parties, facts, issue, relevant sections, evidence status, missing items	Writing a case summary for a lawyer or legal-aid worker

AI techniques. Retrieval-Augmented Generation (RAG) over our own corpus of statute text; multilingual / Bangla-capable text embeddings with a vector database (semantic search); optional cross-encoder re-ranking; LLM integration through an API with constrained prompts and structured (JSON) output for issue, fact and claim extraction; prompt-based claim-versus-statute comparison and contradiction detection supported by simple rule checks for dates and amounts; template-based generation for drafting; and Bangla language AI (Bangla question understanding and Bangla answer generation). Evaluation will use retrieval metrics (Hit@k, MRR), answer faithfulness checks, and manually labelled sample documents for the analysis features.

Why AI is essential. If the AI components are removed, the system cannot match an everyday Bangla question to the correct legal sections, cannot explain them in simple language, cannot extract issues and claims from a document, and cannot compare a document with the law, which is its main purpose; only a plain list of Acts would remain, equivalent to the current portal. The semantic retrieval, the grounded generation and the extraction and comparison steps are the core of the product, not an add-on.

2. System Requirements

This section specifies what the system is expected to do and how well it should perform. Each requirement is written as a testable statement.

2.1 Functional Requirements

ID	Feature	Requirement
FR1	F1	The system shall accept a question typed in Bangla through a web interface.
FR2	F2	The system shall retrieve the top-k (default k = 5) most relevant statute sections for a question.
FR3	F1	The system shall generate an answer in simple Bangla using only the retrieved sections.
FR4	F3	The system shall display, for every answer, the Act name, section number, and original section text it is based on.
FR5	F3	The system shall verify that every section cited in an output is among the retrieved sections, and shall block outputs that fail the check.

ID	Feature	Requirement
FR6	F2	The system shall display a 'cannot find a supported answer' message when the best retrieval score is below a configured threshold.
FR7	F2	The system shall allow the user to filter the search by Act (optional).
FR8	F4	The system shall extract, from a question or document, the legal issue category, key facts, parties, dates and amounts, and display them for the user to confirm or correct.
FR9	F5	The system shall allow the user to upload a text-based PDF, DOCX or TXT file (up to 10 pages / 5 MB) or paste text.
FR10	F5	The system shall show a Bangla summary of the key clauses, dates, amounts and claims found in the uploaded document.
FR11	F5	The system shall tell the user clearly when a file cannot be read (for example a scanned image) instead of producing an analysis.
FR12	F6	The system shall compare each extracted claim or clause with the retrieved sections and label it as consistent, conflicts with, or not covered, with the cited section.
FR13	F6	The system shall state that verification means comparison with the statute snapshot and does not confirm that a document is genuine or that the facts are true.
FR14	F7	The system shall flag possible contradictions between dates, amounts or statements inside a document or case description, and show the conflicting text side by side.
FR15	F7	The system shall flag possible conflicts between a document and the statute text, and mark every flag as 'possible' for user review.
FR16	F8	The system shall compare the user's input with a stored checklist for the identified issue and list the items found and the items not found.
FR17	F9	The system shall provide at least three Bangla draft templates (for example unpaid-wage complaint, consumer complaint, online-harassment complaint) and fill them using the user's details.
FR18	F9	The system shall mark missing details in a draft with clear placeholders, include cited sections, and allow the draft to be edited and downloaded.
FR19	F10	The system shall list next steps only when they are stated in the retrieved sections; otherwise it shall refer the user to a qualified lawyer or legal-aid service.
FR20	F11	The system shall display an evidence status (supported / partly supported / not supported) with a short reason for every answer and analysis output.
FR21	F12	The system shall show related sections from other selected Acts for a retrieved section.
FR22	F12	The system shall show up to three similar example scenarios from the curated scenario library, labelled clearly as examples and not real court cases.
FR23	F13	The system shall produce a mapping table that links each extracted fact to the retrieved section that applies to it, with a citation, or marks the fact as 'no applicable section found'.

ID	Feature	Requirement
FR24	F13	The system shall let the user list the evidence they have (for example appointment letter, payslip, receipt, screenshot, message) by selecting from a list or typing, and shall link each item to the related fact.
FR25	F14	The system shall generate an evidence matrix in which each row is a legal point or fact and each column is an evidence item, with cells marked provided, missing or unclear.
FR26	F14	The system shall allow the matrix to be downloaded as a table (DOCX or CSV) and shall state that it records only what the user has provided.
FR27	F15	The system shall generate a structured Bangla case summary containing parties, key facts, legal issue, relevant sections with citations, evidence status and missing items.
FR28	F15	The system shall build the case summary only from the user's input and the system's verified outputs, and shall allow it to be edited and downloaded.
FR29	All	The system shall show a legal-information disclaimer with every output.
FR30	All	The system shall display the date of the statute snapshot used.
FR31	All	The system shall allow the user to mark an output as helpful or not helpful and optionally add a comment.
FR32	All	The system shall store questions and feedback without personal identifiers, for evaluation only; uploaded documents shall be deleted when the session ends.
FR33	All	An administrator shall be able to add or update the statute data, templates, checklists and scenario library and rebuild the index using a script.
FR34	All	The system shall include an evaluation script that reports Hit@k, MRR, faithfulness and the results of the document-feature tests on the test sets.

2.2 AI Requirements

ID	Feature	Requirement
AIR1	F2	The retrieval component shall return a correct relevant section within the top 5 results for at least 75% of questions in the test set (target; to be reported honestly after evaluation).
AIR2	F3	At least 90% of the section citations in generated outputs shall refer to sections that exist in the retrieved context (faithfulness target).
AIR3	F6, F11	The system shall refuse, or state insufficient information, for questions in the test set that are out of scope (for example unrelated topics or un-indexed Acts) in at least 80% of cases.
AIR4	F1	The generated answers shall be in Bangla and not exceed about 200 words, excluding the quoted section text.
AIR5	All	The system shall not give instructions such as 'file a case against X' or predict outcomes; the prompt and an output check shall enforce a neutral, informational style.
AIR6	F4	The issue category shall be correct for at least 80% of the labelled test questions and sample documents.

ID	Feature	Requirement
AIR7	F5, F6	For the labelled sample documents, at least 70% of key dates and amounts shall be extracted correctly, and at least 70% of claim labels (consistent / conflicts / not covered) shall match the manual labels (target).
AIR8	F7	On sample documents with planted contradictions, the detector shall find at least 60% of them; false alarms shall be reported (target, to be reported honestly).
AIR9	F8	At least 80% of the checklist items marked missing by the system shall match the manual judgement on the sample documents.
AIR10	F9, F10	Every draft and every next-step list shall contain only sections that are in the retrieved set, and shall be reviewed by the team against the source text on a sample of at least 20 outputs.
AIR11	F11	Outputs labelled 'supported' shall have a higher manual-faithfulness rate than outputs labelled 'partly supported' or 'not supported' on the test set.
AIR12	F12	For at least 70% of test questions, a human judge shall consider at least one of the top 3 related provisions or scenarios relevant (target).
AIR13	F13	On the labelled sample cases, at least 70% of extracted facts shall be linked to a correct section, and facts with no applicable section shall be marked as such rather than forced to a section (target).
AIR14	F14	At least 75% of the evidence-matrix cells (provided / missing / unclear) shall match the manual labels on the sample cases (target).
AIR15	F15	Case summaries shall include the key parties, dates, amounts and issue in at least 80% of sample cases, and a manual check of at least 20 summaries shall find no statement that is not supported by the input or the cited sections.

2.3 Non-Functional Requirements

- Performance: a typical answer shall be shown within 10 seconds, and a document analysis of up to 10 pages within 60 seconds, on a normal internet connection.
- Usability: the interface shall be simple, support Bangla fonts correctly, and be usable on mobile browsers.
- Privacy: no names, phone numbers or account data shall be collected; users are warned not to enter or upload personal details; uploaded files are processed in the session and not kept.
- Maintainability: data preparation, indexing, templates, checklists and evaluation shall be separate scripts or files so that statute data can be refreshed.
- Transparency: the sources, the evidence status and the statute snapshot date shall always be visible.
- Reliability: if the LLM API or a file reader fails, the system shall show a clear error message and shall not show an unverified answer.
- Security: API keys shall be kept in environment variables and not in the source code; uploaded file types and sizes shall be validated.

3. Tools and Technologies

Category	Tools (proposed)
Programming languages	Python (backend, AI pipeline, data processing); JavaScript/HTML/CSS (frontend)
Backend framework	FastAPI
Frontend	Streamlit for the prototype, or React for a custom interface (final choice by team skill)
Data collection and processing	Official Bangladesh Laws portal as source; Requests/BeautifulSoup for collection (respecting terms of use); PDF text extraction libraries where needed; regular expressions and Pandas for cleaning and section-wise splitting
Document handling (F5)	pypdf or pdfplumber for text-based PDFs; python-docx for DOCX; plain-text reader; python-docx for exporting drafts
Databases	ChromaDB or FAISS (vector store); SQLite (feedback and logs); JSON/CSV files for templates, checklists and the scenario library
Embedding models	Open multilingual embedding model with Bangla support (e.g., multilingual-e5 or BGE-M3 via Sentence-Transformers); the best one will be selected by testing on our question set
Language model	A Bangla-capable LLM through an API (e.g., Claude API or another provider); prompt-constrained to retrieved context; JSON-formatted output for extraction
Re-ranking (optional)	Multilingual cross-encoder re-ranker
RAG orchestration	LangChain or LlamaIndex, or a small custom pipeline
Evaluation	Own test set (CSV) and labelled sample documents; Hit@k, MRR; manual faithfulness check plus a model-based judge as a supplement
Version control and collaboration	Git and GitHub; Google Drive for documents
Deployment (demo)	Local machine or a free-tier cloud service (e.g., Hugging Face Spaces or Render)

4. Project Timeline and Work Plan

The plan assumes a 14-week semester and a team of three members; durations will be adjusted to the actual course calendar. To keep the fifteen features realistic, they are grouped in priority tiers. Tier 1 (F1-F4, F11) is the core pipeline and is completed first. Tier 2 (F5, F6, F8, F10, F13) builds on the same retrieval and prompts. Tier 3 (F7, F9, F12, F14, F15) is implemented in a limited form (for example three templates, a small scenario library, a matrix limited to the main legal points) and reduced further if the schedule is under pressure.

4.1 Milestone Table

Week	Milestone	Main tasks	Deliverable
1-2	M1: Planning and data selection	Finalise list of Acts; study text availability; set up repository; define question-set and sample-document formats; fix feature tiers	Proposal, repository, Act list
3-4	M2: Knowledge base ready	Collect and clean statute text; split by section; add metadata; start issue checklists and procedure notes	Clean section-wise dataset
5-6	M3: Retrieval prototype	Compare embedding models; build vector index; test retrieval (F2); related-provision lookup (F12)	Working semantic search
7-8	M4: RAG answer generation (Tier 1)	Design prompts; integrate LLM; Q&A (F1); issue extraction (F4); citations and verification (F3); refusal rule and evidence status (F11)	End-to-end backend for Tier 1
9-10	M5: Document and case intelligence (Tier 2)	Document upload and reader (F5); claim comparison (F6); missing-information checker (F8); next-step guide (F10); fact-law-evidence mapping (F13)	Document analysis working
11	M6: Tier 3 features	Contradiction detector (F7); Bangla templates and drafting (F9); scenario library and finder (F12); evidence matrix (F14); case summary generator (F15)	All 15 features at prototype level
12	M7: Interface and integration	Build web UI for all features; connect backend; feedback and disclaimer	Working demo
13	M8: Evaluation and improvement	Finalise 100+ question test set and sample documents; run metrics; error analysis; tune thresholds and prompts; user testing with classmates	Evaluation results
14	M9: Final submission and demo	Bug fixing; final report; presentation and demo; code cleanup	Final report, demo, source code

4.2 Task Distribution Among Team Members

Member	Responsibilities
Member 1 (Data and retrieval)	Statute collection, cleaning, section splitting, metadata and snapshot documentation; embedding model comparison, vector database, re-ranking, retrieval evaluation; F2 (RAG retrieval), F12 (similar provision / scenario finder) and F15 (case summary generator); scenario library
Member 2 (Generation and analysis)	Prompt design and LLM integration; F1 (Q&A), F3 (citation verification), F4 (issue extraction), F6 (verification engine), F7 (contradiction detector), F11 (evidence status), F13 (fact-law-evidence mapping); refusal logic and safety checks

Member	Responsibilities
Member 3 (Interface, documents and evaluation)	Frontend and API integration; F5 (document upload and analysis), F8 (missing-information detector), F9 (drafting templates), F10 (next-step guide), F14 (case evidence matrix); feedback storage; test questions and sample documents; user testing
All members	Documentation, weekly meetings, code review, cross-checking of statute text used in templates and checklists, presentation

5. Additional Sections

5.1 Existing System

Currently, people use (a) the official laws portal for browsing and keyword search, (b) general web search and social-media groups, and (c) general chatbots. These are free and available, but they do not combine Bangla question understanding, exact statute citations, document checking, and verification. Our system differs by restricting answers to a verified statute snapshot, showing the source text beside the explanation, showing how well each output is supported, and extending the same evidence to documents, drafts and next steps.

5.2 System Architecture

The system has four layers: (1) Data layer - cleaned statute sections and metadata, a vector index, draft templates, issue checklists, the scenario library, and a small database for feedback; (2) AI layer - embedding model, retriever and re-ranker, LLM generator and extractor, claim comparator, and citation verifier; (3) Application layer - FastAPI services that orchestrate the pipeline and the document reader; (4) Presentation layer - a web interface that shows the output, sources, evidence status and disclaimer. Figure 1 shows the flow and marks the AI components.

5.3 Ethical, Legal, and Social Considerations

- Not legal advice: every output carries a disclaimer and a suggestion to consult a qualified lawyer or legal-aid service for real cases.
- Accuracy and hallucination risk: grounding, citation checking, evidence status and refusal behaviour reduce, but do not remove, the risk of errors; the limitations will be stated openly and evaluation results reported honestly.
- Meaning of 'verification': the verification engine compares text with the statute snapshot; it does not authenticate documents or prove facts, and this is stated in the interface. Contradiction and missing-information flags are shown as 'possible' so users do not treat them as final findings.
- Evidence matrix and case summary: the matrix records only what the user has provided and does not judge the strength or genuineness of evidence; the case summary is generated only from the user's

input and verified outputs, is shown with its evidence status, and must be reviewed by the user (and a lawyer for real cases).

- Drafting risk: drafts are templates for the user to review; the system does not file anything and warns users to check details before use.
- Outdated law: the knowledge base is a dated snapshot; amendments after that date may not be included, and the date is displayed.
- Privacy: no personal data is required; questions are stored without identifiers; uploaded documents are processed only for the session and deleted afterwards; users are advised not to enter or upload names or sensitive details.
- Bias and language fairness: performance may be weaker for dialectal or very informal Bangla; the test set will include varied phrasing to measure this.
- Data source and attribution: official texts will be used with proper attribution and in line with the source's terms of use; the project is non-commercial and educational.
- Sensitive topics: for topics such as domestic violence, the output will also show relevant support-service information in a neutral and careful way.

5.4 Future Work

- Add more Acts, rules, and later, court judgments (to turn the example-scenario finder into a real case finder).
- OCR support for scanned documents and voice input and output for users with low reading ability.
- Automatic detection of amendments and repealed sections.
- Fine-tuning a Bangla embedding model on legal question-section pairs.
- More draft templates and lawyer-reviewed templates, checklists and answer sets, and a community feedback loop.
- A mobile application and offline support.